

VLMS

A C class IP address 192.168.10.0 I have selected to create subnets for five (5) computer labs with fourteen usable ips.

So CIDR for main IP is 192.168.10.0/24 and its subnet Mask is 255.255.255.0

For 12 useable ips I will borrow four (4) **Msb** bits for network in last useable octet.

$$2^4 = 16$$

New mask $256 - 16 = 240$

New CIDR = **192.168.10.0/28**

Full New Subnet Mask = **255.255.255.240**

Network IP	Broadcast IP	Stating IP	Ending IP
192.168.10.0	192.168.10.15	192.168.10.1	192.168.10.14
192.168.10.16	192.168.10.31	192.168.10.17	192.168.10.30
192.168.10.32	192.168.10.47	192.168.10.33	192.168.10.46
192.168.10.48	192.168.10.61	192.168.10.49	192.168.10.60
192.168.10.62	192.168.10.77	192.168.10.63	192.168.10.76

Router and Routes

I have used three (3) routers and five switches to create five labs networks.

For Router0 and Router1 connection I have used 192.168.11.0/24 network and for Router1 and Router2 connection I have used 192.168.12.0/24. I have set Static routing for Router0 to Router1 and **Eigrp Dynamic routing** for Router1 to Router2.

Routers configuration commands used are:

- Enable
- Configuration terminal (Conf t)
- Interface Fa2/0 or Se0/0
- Ip address 192.168.10.2 255.255.255.240
- Clock rate 56000 (for serial ports)
- No shutdown
- Exit
- Ip route 192.168.10.32 255.255.255.240 192.168.11.2
- Ip route 192.168.10.48 255.255.255.240 192.168.11.2
- Ip route 192.168.10.62 255.255.255.240 192.168.11.2
- Router eigrp 100

- Network 192.168.10.32
- Network 192.168.11.0
- Network 192.168.12.0
- Redistribute static metric 100
- Exit
- Write Memory